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SECTION 4.0: OPERATING THRESHOLDS & SAFE OPERATING WINDOWS (SOW)

4.1 Scope: Applicable to all API 610 single/multi-stage centrifugal pumps in CDU/VDU.

4.2 Bearing Temperature Limits:

- Normal Continuous Range: 45 deg C to 70 deg C.
- Advisory Warning Threshold: 75 deg C.
- High Trip / Shutdown Mandate: Exceeding 80 deg C requires plant notification and scheduled shutdown within 48 hours to avert catastrophic shaft seizure.

4.3 Vibration Severity Criteria (ISO 10816-3 Group 1/2 Rigid Mount):

- **Zone A (Good): 0.0 - 2.8 mm/s RMS.**
- Zone B (Acceptable): 2.8 - 4.5 mm/s RMS.
- Zone C (Unrestricted Operation Not Permitted / Alarm): > 4.5 mm/s RMS.
- Zone D (Trip / Immediate Danger): > 7.1 mm/s RMS.

4.4 Mechanical Seal Integrity (API 682 Standard):

- Normal barrier fluid leakage: < 2 drops/min.
- Leakage exceeding 5 drops/min indicates primary face degradation.
- Active leakage at 15 drops/min demands cartridge replacement and barrier flush.

4.5 Overhaul Intervals: Full rotating assembly overhaul every 12 months in crude service.